Pooling signed power certificates before integral projection

A direct quartic consistency equation in 218 integer variables, with a 216-variable companion

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Abstract

We give a fully specified polynomial $P_{218} \in \mathbb{Z}[x_0, \ldots, x_{217}]$ of total degree four for which an ordinary argument formalizable in ZFC proves

$$\big(\forall \mathbf{x} \in \mathbb{Z}^{218}\ P_{218}(\mathbf{x}) \neq 0\big) \quad \longleftrightarrow \quad \mathrm{Con(ZFC)}.$$

It is the sum of squares of 155 explicitly printed quadratic residuals. Its collected expansion has 7,726 nonzero monomials and maximum absolute coefficient 45,161. The inspected 225-variable direct equation has 162 residuals, 7,867 monomials, and height 128,446. Thus the new principal equation improves all four size measures, with degree unchanged. A second exact construction has 216 variables and 153 residuals, at the measured cost of 15,158 monomials and height 334,952.

The construction combines the factorial-free arithmetic source of ProveIt source 157 with the fixed-base power pooling of source 158, then applies explicit integral projections and residual row operations. The combination requires a new check of the signed bootstrap and of every retained square-frame atom. Every constant of the final polynomials is a literal integer. There is no hidden universal-polynomial parameter or Turing-machine coefficient tuple.

The inherited C222/ZF/ZFC refutation calibration is identified exactly. The new semantic composition and algebraic maps are ordinary proofs, with independent finite coefficient checks. This is a concrete improvement in the direct quartic branch, not a global simplicity minimum, a consistency decision, or a newly compiled end-to-end formal proof.

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1 The delivered equation and the scope of the result

All independent unknowns of the delivered equations range over *unrestricted integers*. Let $R_0, \dots, R_{154}$ be the literal quadratic polynomials printed in Appendix A. The principal equation is

$$\boxed{P_{218}(x_0, \dots, x_{217}) = \sum_{j=0}^{154} R_j(x_0, \dots, x_{217})^2 = 0.} \tag{1}$$

The appendix, together with this finite sum, completely defines one polynomial. The machine-readable coefficient list and complete collected expansion are supplied as well. The indices of the sum are fixed; they are not additional unknowns.

Theorem 1.1 (Consistency equation)**.** *For the effective presentation of ZFC used by the repository's named refutation calibration, ZFC proves*

$$\exists \mathbf{x} \in \mathbb{Z}^{218}\ P_{218}(\mathbf{x}) = 0 \quad \longleftrightarrow \quad \exists p\ \mathrm{Refutation}_{\mathrm{ZFC}}(p).$$

Consequently ZFC proves that (1) *has no integer solution if and only if* Con(ZFC)*. The analogous statement holds for the supplied* $P_{216} \in \mathbb{Z}[x_0, \dots, x_{215}]$*.*

The theorem composes the arithmetic equivalences and integer polynomial maps below with the inherited syntactic calibration. It asserts neither side of the consistency equivalence by itself. Evaluating these polynomials is not a method for deciding consistency.

The principal residual height is 128, and all 218 declared coordinates occur. The companion residual height is 384. The companion uses two fewer coordinates but is less sparse and has larger coefficients.

The comparison is pinned to ProveIt commit `953fb4d82d4229f4efc5f9db4cc2bf38ae329bb4`. The repository also has compact equations with far fewer variables and much higher degree or very large fixed coefficients. At that snapshot its complete display uses 15 strictly positive variables, 66 integer-leaf binary operations, and degree 157 [6]. That result has a different domain and objective. Our contribution is an explicit improvement in the direct, small-coefficient, degree-four construction.

Equation	Variables	Rows	Degree	Terms	Height
Source 158, P_{225}	225	162	4	7,867	128,446
New principal P_{218}	218	155	4	7,726	45,161
New companion P_{216}	216	153	4	15,158	334,952

Table 1: Exact counts over $\mathbb{Z}$. Rows are quadratic residuals before taking the sum of squares. Height is the maximum absolute *expanded polynomial* coefficient.

A final read-only check found sources 160–165 at commit `fb1bed5eb7196cf35abfaa746c18c6ee7eae6cfb` [7]. Their READMEs report further tradeoffs: source 162 has a 236-variable quartic with 4,876 monomials and height 7,968; source 165 has a 238-variable quartic with 5,236 monomials and height 3,968. Our two forms use fewer variables, while those examples have smaller support and height. The simultaneous improvement in Table 1 is specifically over the pinned source 158 equation. Later metrics were read from the cited READMEs, not independently expanded. Their exact file identities are recorded in `provenance/LATE_REPOSITORY_CHECK.json`.

2 Logical calibration and the starting arithmetic system

2.1 The inherited chain

The arithmetic source encodes finite derivations in the numerical calculus C222, rather than a particular Turing-machine execution. Its logical chain is

$$\begin{aligned} \text{integer zero} &\longleftrightarrow \text{finite square-shell C222 refutation} \\ &\longleftrightarrow \text{finite original C222 refutation} \\ &\longleftrightarrow \text{finite ZF refutation} \\ &\longleftrightarrow \text{finite ZFC refutation.} \end{aligned} \tag{2}$$

The first equivalence is arithmetic; the second is a recursive recoding of formula constructors. The last two use the declarations

```
KEndpoint.c222_iff_zf_refutation
ConstructibleChoice.zf_refutation_iff_zfc_refutation.
```

These are unconditional calculus/refutation interfaces [4, 5]; machine-specific simulation hypotheses are not used.

We retain the source statements and their pinned identities. We have not rebuilt their entire transitive Lean dependency graph. The ordinary proofs used here can be carried out in ZFC; this report does not supply an encoded object-ZFC derivation or a fresh full Lean or Rocq proof term. This is the dependency boundary of Theorem 1.1.

2.2 The syntax being arithmetized

For $a, b \in \mathbb{N}$ and $s = a + b$, the square-shell recoding uses

$$Q_0(a, b) = s^2 + a, \qquad Q_1(a, b) = s^2 + 3a + 4b + 3$$

and four constructors

$$I = 2Q_0, \quad M = 2Q_0 + 1, \quad A_{\text{syn}} = 2Q_1, \quad E_{\text{syn}} = 2Q_1 + 1.$$

The two shell ranges partition the natural numbers before the parity split. Recursive decoding yields a bijection with the original coding. It fixes zero and the falsity code one. Constructor identities and freshness disequalities transport through this bijection. The source 158 syntax note identifies this interface [2].

The local compiler represents a backwards proof-tree expansion by a finite list of polynomial alternatives. A pending multiset of labels is represented by a product of weights $1 + jF$. Local transitions replace a conclusion by zero, one, or two premises. A beta/CRT code stores the accumulator path; binomial divisibility selects its rows and enforces bounded local parameters. Pell certificates replace all requested powers by ordinary polynomial equations.

These mechanisms are inherited. The present construction preserves the exact local square-shell Horn circuit and changes only the identified range and certificate layers.

2.3 The factorial-free source

Our actual starting system is the 242-coordinate arithmetic source in package 157, before that package's later projections [1]. Its literal file is a pinned input. Its semantic quantities include

$$\begin{gathered} t = 16 + 4d^2, \qquad F = t^t, \qquad C = 2c, \\ H = 2 + 4(C + A_1 + \cdots + A_8), \qquad y = F(2v + 1), \\ Bh_B = C + F + 1, \\ x = F + 1 + (y + 1)u_0, \qquad x = 1 + [1 + (t + 1)y]u_t. \end{gathered} \tag{3}$$

Exponentiation in this display describes the meanings proved for polynomial certificates; it is not a primitive of the final equation. The initial requests are $F = t^t$, $N = B^x$, $q = (BH)^x$; the other requests use the two bases q and $q + 1$. The binomial radius is F, with authenticated digit $B = \binom{F+t}{t}$. Authenticating $t!$ is unnecessary.

The retained square frame has

$$W = 2 + \sum_{\nu} a_\nu^2, \qquad W \geq 258, \qquad |A| > W^W. \tag{4}$$

Every actual requested base, index, and proposed output is among the affine atoms a_ν. The retained orientation argument identifies the Pell pairs at their canonical *signed* indices before the arithmetic signs in (3) are known. The distinction between $a_\nu^2 < W$ and the weaker $|a_\nu| < W$ is essential for pooling. Independent checkers reconstruct the actual 36-square identity and check all 17 retained power requests.

3 Cofinal strict ranges

Five old tests use lower binomial index $t + 1$: the four local schema parameters and the derived label. We replace all five by tests at lower index t. The decoded bounds become

$$0 \leq a, b, c, d < t, \qquad 0 \leq J(a, b, c, d) < t. \tag{5}$$

This is a bound on each *local* tuple. We do not bound the much larger global value of J on the CRT representatives by t; that global value remains the semantic Pell anchor.

Proposition 3.1. *The strict-range replacement preserves existence of an integer solution of the factorial-free arithmetic source.*

Proof. Soundness uses the inherited prime-power range lemma: if $p^e \mid \binom{a}{k}$, $a \geq 0$, $k, e \geq 1$, then the least residue of a modulo p^e is less than k. Taking $k = t$ strengthens the decoded local bounds, so previous soundness conclusions continue to hold.

For completeness fix a finite refutation tree. Choose $t = 16 + 4d^2$ strictly larger than its number of expansion steps, every local parameter, and every value of the full local derived-label expression, including inactive branches and the truth-padding tuple. This is a cofinal choice made before constructing F, the accumulators, the beta code, or the Pell frame.

For each full prime power $p^m \parallel B$, hence $p^{xm} \parallel N = B^x$, precision-$2x$ CRT makes a representative congruent to its target $w \in [0, t-1]$ modulo p^{2xm}. If the representative is below t, its binomial at lower index t is zero. Otherwise its falling factorial contains $a - w$. Since

$$v_p(t!) \leq t - 1 < xm,$$

division by $t!$ leaves divisibility by p^{xm}. Polynomial congruence gives the same conclusion for the derived label. The factorial-free source supplies the much stronger endpoint bound $x \geq 2 + (t+1)y$, so replacing an early ratio consequence $x \geq t + 1$ by $x \geq t$ causes no loss. □

Every affected normalized digit can now use the existing q^t denominator. Delete the q^{t+1} output, its private power quotient, the wire for its derived Pell pair, and its normalization ratio. Delete their three rows, redirect five ratio uses, and remove the old output square from the frame. The system has 238 coordinates, and all principal orientation pairs remain.

This is a semantic transformation: an old witness need not survive at its old horizon. Completeness rebuilds witnesses at a larger horizon. We do *not* use source 158's separate factorial-dependent change $y = V - 1$; here $y = F(2v + 1)$ is retained.

4 Two quadratic rows for a whole block of powers

The following fixed-base construction is inherited from source 158. Its signed hypotheses and its use in the new bootstrap are checked again for this combined source.

Lemma 4.1 (Signed fixed-base pooling)**.** *Let $W \geq 258$, $|A| > W^W$, and $1 \leq r \leq 15$. Let $n \neq 0$, and suppose, for each $1 \leq i \leq r$,*

$$n^2 < W, \qquad m_i^2 < W, \qquad k_i^2 < W.$$

Let (X_i, Y_i) be the canonical signed Pell pair at parameter A and integer index k_i, and put $Z_i = AY_i$. There exist $K, L \in \mathbb{Z}$ satisfying

$$K - 2\sum_i Y_i(m_i - X_i) - (2A - n)L = 0, \tag{6}$$

$$\sum_i [(m_i - X_i + Z_i)^2 - Y_i^2] - nK + L = 0 \tag{7}$$

if and only if every exact power identity $m_i = n^{k_i}$ holds. For $|n| \geq 2$, these identities force $k_i \geq 0$. For $n = \pm 1$, signed indices are allowed.

Proof. Write

$$T = 2An - n^2 - 1, \quad D_i = m_i - X_i + Z_i, \quad U = \sum_i Y_i(m_i - X_i), \quad S = \sum_i (D_i^2 - Y_i^2).$$

The polynomial identity

$$\sum_i (D_i - nY_i)^2 = S - 2nU - T\sum_i Y_i^2 \tag{8}$$

holds when $Z_i = AY_i$. Eliminating K from the two rows gives $S - 2nU = TL$. Thus T divides the left side of (8).

The Pell congruence remains valid for negative indices. In the ring with formal square root of $A^2 - 1$, sending that root to $n - A$ is a homomorphism modulo T, because $(n - A)^2 - (A^2 - 1) = -T$. It sends the norm-one unit $A + \sqrt{A^2 - 1}$ to n. Since $T \equiv -1 \pmod{n}$, n is invertible modulo T. Hence

$$X_i + (n - A)Y_i \equiv n^{k_i} \pmod{T}$$

for all signed indices.

Put $b = \lfloor \sqrt{W - 1} \rfloor$. Multiplying by n^b clears all negative indices and gives

$$T \mid E := \sum_i (m_i n^b - n^{b+k_i})^2.$$

All exponents are nonnegative. Every $|n|, |m_i|, |k_i| \le b$, with $b \ge 16$, so

$$0 \le E \le 4rb^{4b} < W^{2b+1} \le W^W.$$

Here $4r \le 60 < W$, $b^2 < W$, and $2b + 1 \le b^2 + 1 \le W$. On the other hand,

$$|T| \ge 2|A||n| - n^2 - 1 \ge 2|A| - W > W^W.$$

Thus $E = 0$, so each square vanishes. If $|n| \ge 2$ and $k_i < 0$, the resulting equation $m_i n^{-k_i} = 1$ is impossible over $\mathbb{Z}$. All other cases give the asserted powers.

Conversely the exact powers make each $D_i - nY_i$ divisible by T. Identity (8) then makes $L = (S - 2nU)/T$ integral. Taking $K = 2U + (2A - n)L$ supplies both rows. □

The symbols T, U, S, b are proof abbreviations, not new coordinates. The other quantities are existing affine coordinate expressions, including the existing $Z_i = AY_i$ gate. Thus both new rows are quadratic. A whole block needs just two new coordinates and two rows in this construction.

The strict system has eleven requests at $q + 1$ and three at q. Replacing individual quotients and the old modulus coordinates saves ten and two coordinates, respectively. The initial requests $F = t^t$, $N = B^x$, and $q = (BH)^x$ remain individual.

5 The pooled factorial-free bootstrap

An algebraic pooling identity alone is insufficient. The bases and signed indices must be justified in an order that avoids circular sign assumptions.

Proposition 5.1. *The modified system recovers all arithmetic signs and exact powers without assuming a positive radix or positive CRT representatives.*

Proof. The individual request at known base and index t first gives $F = t^t \ge 16^{16} = 2^{64}$. Since $t = 16 + 4d^2$, F is even. With $C = 2c$, the right side of $Bh_B = C + F + 1$ is odd even when $c < 0$. Thus B is odd and nonzero. Independently $H \equiv 2 \pmod{4}$, whence $|H| \ge 2$ and $|BH| \ge 2$. The individual BH request gives $x \ge 0$ and $q = (BH)^x$.

The beta equation gives $|y| \geq F$, so

$$|1 + (t+1)y| \geq (t+1)F - 1 > 1.$$

If $x = 0$, the terminal endpoint makes this integer divide -1, a contradiction. Therefore $x \geq 1$ and $|q| \geq 2$.

Apply Lemma 4.1 to the q block. Its request at exponent $y+1$ forces $y+1 \geq 0$. A negative y would be at most $-F < -1$, so $y \geq F > 0$. The first endpoint excludes $u_0 \leq -1$, giving $x \geq F+1$; the last endpoint now forces $u_t \geq 1$. Thus

$$x \geq 2 + (t+1)y \geq 2 + (t+1)F. \tag{9}$$

Since $|q| \geq 2^x$, we have $|q+1| \geq 2$. Pooling at $q+1$ now proves all representative indices nonnegative and all outputs exact. In particular $C, A_1, \ldots, A_8 \geq 0$, the semantic anchor retains its meaning, and $H > 0$. Neither $B > 0$ nor $q > 0$ has yet been assumed.

To finish the sign argument, write $Q = |q|$ and $c_j = \binom{F+t}{j}$. For $0 \leq j \leq t$,

$$c_j \leq (F+t)^t \leq (F^2+F)^t \leq 2^{t(F-1)}.$$

Equation (9) yields $c_{\max}/(Q-1) < 1/32$ and $c_t < Q/32$. The binomial theorem is valid at negative q, and its true low part $r_0 = \sum_{j<t} c_j q^j$ satisfies $|r_0| < Q^t/32$.

The retained normalization identities and common remainder norm bound each supplied remainder by half its denominator in absolute value. Congruent supplied and genuine remainders must therefore coincide. The digit row gives $B \equiv c_t \pmod{q}$. For $a = |BH| \geq 2$, $|B| \leq a < Q/2$, because $x \geq 3$. Hence $|B - c_t| < Q$, and

$$B = c_t = \binom{F+t}{t} > 0.$$

Now $q > 0$, $N = B^x > 0$, and all later binomial, CRT, and weight arguments have their original natural-number meanings. □

Source 157's remaining estimates are unchanged. They use $\operatorname{lcm}(1, \ldots, t)^2 \leq 16^{t-1} < t^t = F$ for binomial CRT selection, its sharper weight-separation proof at $F = t^t$, and its local polynomial majorant below 2^x [1]. The local ranges have only become smaller.

For reverse construction choose the larger cofinal horizon, choose the beta spacing as a sufficiently large odd multiple of the odd part of $t!$ times F, then build the original CRT and simultaneous Pell witnesses. The converse of Lemma 4.1 supplies the four pooled coordinates. Old witnesses need not remain numerically unchanged.

6 Exact algebraic compression

6.1 Private wires and Pell difference coordinates

Before pooling, three private coordinates are removed by the consumer-based integer changes of source 158. Their forward and reverse residual identities are checked against the factorial-free source by matching the unchanged local Horn subcircuit. This takes 238 coordinates to 235. The change introduces one shared product, eliminates four private wires, and replaces a gate by an aggregate square-sum coordinate. It is an exact algebraic transformation, unlike the cofinal change of Section 3.

Pooling gives 223 coordinates and 160 residuals. For each retained Pell pair use the difference coordinate

$$U_i = X_i - Z_i, \qquad X_i = U_i + Z_i.$$

Because Z_i already has a gate coordinate, this is an invertible integer affine change in the ambient coordinate system. On the common zero set, the retained gate equation $Z_i = AY_i$ identifies U_i with $X_i - AY_i$. It reduces expanded support without changing the number of unknowns or requiring a divisibility premise. The resulting 223-coordinate system is the pinned input to the following projection theorem.

6.2 Removing common quadratic directions

Theorem 6.1 (Integral radical projection)**.** *Let $F_0, \dots, F_{m-1} \in \mathbb{Z}[\mathbf{x}]$ have degree at most two. Suppose $v \in \mathbb{Z}^n$ satisfies*

$$F_i(\mathbf{x} + uv) = F_i(\mathbf{x}) + ua_i \quad \text{for every } i,$$

where $a_i \in \mathbb{Z}$ is independent of $\mathbf{x}$. Assume $v_p = 1$ and $a_r = \pm 1$. On $z_p = 0$, set

$$G_i(\mathbf{z}) = F_i(\mathbf{z}) - (a_i/a_r)F_r(\mathbf{z}) \qquad (i \neq r).$$

The common integer zero sets of F and G are in polynomial bijection, and every G_i has degree at most two.

Proof. The projection and lift are

$$\pi(\mathbf{x}) = \mathbf{x} - x_p v, \qquad \lambda(\mathbf{z}) = \mathbf{z} - vF_r(\mathbf{z})/a_r.$$

They have integer coefficients because $a_r = \pm 1$. The directional identity gives

$$F_r(\lambda(\mathbf{z})) = 0, \qquad F_i(\lambda(\mathbf{z})) = G_i(\mathbf{z}).$$

At a zero of F, $F_r(\pi(\mathbf{x})) = -a_r x_p$, and therefore $\lambda(\pi(\mathbf{x})) = \mathbf{x}$. Conversely $z_p = 0$ gives $\pi(\lambda(\mathbf{z})) = \mathbf{z}$. □

A common kernel vector for the homogeneous Hessian matrices has the required constant-direction property. The certificate supplies five integer vectors, pivots, omitted rows, and directional constants. The checker verifies the directional identities themselves, not merely a reported rank.

Dimension	Pivot	Omitted row	Nonzero entries of v
223	43	28	$v_{43} = 1,\ v_{44} = -1$
222	59	45	$v_{59} = 1,\ v_{60} = -1,\ v_{61} = -2$
221	63	48	$v_{61} = -1,\ v_{63} = 1$
220	30	15	$v_{30} = 1,\ v_{31} = -1,\ v_{32} = -2$
219	24	9	$v_{24} = 1,\ v_{26} = -1,\ v_{27} = -2$

Table 2: The five actual radical projections. Indices refer to the current system before that row of the table is applied.

The transformations take 223 coordinates and 160 rows to 218 coordinates and 155 rows. This is source 159's method applied to the new combined arithmetic source [3]; the generic projection theorem is not claimed as a new literature theorem.

Then 38 recorded integer row operations lower height and support. Each $F_i \leftarrow F_i + kF_j$, $i \neq j$, has integer inverse $F_i \leftarrow F_i - kF_j$, so the common zero set is unchanged. The sum of squares of the final rows is P_{218}.

6.3 The 216-coordinate companion

Two affine consequences permit two additional coordinate eliminations. Each certificate expresses an affine polynomial as an exact constant-coefficient combination of the rows, with unit coefficient on the removed coordinate. Solving for that coordinate and substituting preserves degree two. The certificate identifies the omitted old row and proves its recovery from the combination. The row-combination coefficient need only be nonzero for this recovery over $\mathbb{Z}$; the solved coordinate coefficient must be a unit to ensure an integral lift.

Direct substitution needlessly enlarges coefficients. In the 218-coordinate projected system before the balancing row operations, let

$$a = x_8, \quad b = x_9, \quad u = x_{19}, \quad v = x_{20}, \quad w = x_{33}.$$

The exact identity is

$$-F_5 + F_{18} = 4 + 12a + 16b + 4u - v + w.$$

Introduce the integral affine coordinates

$$t_* = u + 3a + 4b + 1, \qquad c_* = v - 2t_*,$$

whose inverse is

$$u = t_* - 3a - 4b - 1, \qquad v = c_* + 2t_*.$$

The affine relation becomes $2t_* - c_* + w = 0$, hence $w = c_* - 2t_*$. The retained common-square row collapses to

$$c_* - 4t_*^2 + 1 = 0.$$

No integer coordinate is divided by two. The actual chart and inverse are explicit certificate data, and the notation here does not add polynomial unknowns.

The two affine eliminations, followed by 59 invertible row operations, yield P_{216}. Its 153 quadratic residuals and full collected expansion are included. The centered chart controls the coefficient growth caused by the two additional substitutions. The principal P_{218} remains the better sparse, low-height choice.

7 Proof of the main theorem

Proof of Theorem 1.1. A finite sum of integer squares is zero precisely when every residual is zero. Thus a zero of P_{218} is a zero of its 155-row quadratic system.

Invert the recorded row operations and apply the lifts of Theorem 6.1, obtaining a 223-coordinate zero. Invert the Pell difference chart. The actual square-frame memberships, Lemma 4.1, and the bootstrap of Proposition 5.1 restore every exact power. The exact private-wire maps and the strict-range soundness argument recover the inherited local proof-certificate interpretation. It gives a finite square-shell C222 refutation. The recoding and named interfaces in (2) produce a finite ZFC refutation.

Conversely a finite ZFC refutation yields a finite C222 refutation and its square-shell recoding by the inherited calibration. Choose a sufficiently large cofinal horizon and construct the factorial-free strict-range witnesses, as in Section 3 and source 157's completeness proof. The private-wire changes, the converse of Lemma 4.1, and the Pell difference chart give a 223-coordinate zero. Apply the radical projections and recorded row operations to obtain a zero of P_{218}.

All steps are finite integer arithmetic, recursive syntactic constructions, or the identified inherited calibration, and their arguments formalize in ZFC. Negation gives the stated consistency equivalence. The additional centered chart and affine projections give the same equivalence for P_{216}. □

8 Finite verification and reproducibility

Verification separates three claims. The article and independent mathematical review establish the new unbounded semantic steps using the identified inherited arithmetic theorems. The composition checkers bind those statements to the literal source edits, all actual power requests and the square frame, and the private coordinate maps. The algebra checker verifies every projection, row operation, inverse chart, and collected coefficient. Coefficient checks do not replace unbounded proofs.

The root commands use only Python's standard library:

```
python3 -B verify.py
python3 -B -O verify.py
```

Both ordinary and optimized execution are supported; required checks are not removable Python assertions. The receipts record the checks actually performed. Deliberately corrupted certificates are rejected. There is no search for a zero.

The sparse format stores each term as an integer coefficient and a sorted list of coordinate indices. Repetitions mean repeated factors; the empty list denotes the constant monomial. For example $[-3, [2, 7, 7]]$ denotes $-3x_2x_7^2$. Monomial index lists in residuals have length at most two, and in the expanded quartic at most four. Validation excludes duplicate monomials, zero coefficients, noninteger data, and out-of-range indices.

The verifier independently expands $\sum R_j^2$, compares every coefficient with the collected polynomial, and checks degree, support, height, coordinate liveness, and the printed principal appendix. All 155 residuals in that appendix are parsed back into integer polynomials for comparison. Raw source identities and canonical JSON hashes are recorded separately.

The complete literal equations can be evaluated without executing a historical compiler. The builder and frozen certificates are also supplied for reproduction. The following table gives the construction stages; changes of integer basis leave the counts of variables and rows fixed.

Stage	Coordinates	Quadratic rows
Pinned factorial-free arithmetic source	242	178
Strict local ranges	238	175
Private-wire basis	235	172
Pooled powers and Pell differences	223	160
Common-direction projection	218	155
Centered affine projections	216	153

8.1 Attribution and verification boundary

The new result is the complete cross-branch construction, its checked noncircular bootstrap, the exact finite bindings, and the two final literal polynomials. The factorial-free binomial/CRT theorem is inherited from source 157; the pooling lemma and private-wire identities from source 158; the common-direction projection method from source 159. The combined instance and its improved equations are the contribution of this iteration.

New finite checks were executed for this release. Historical source receipts remain historical. The full logical formal development was not recompiled. No global optimum or literature-wide priority is asserted, and the internal independent reviews are AI-assisted audits, not external human peer review.

9 Further research questions

1. Can a residual basis for the centered 216-coordinate companion approach the support and height of P_{218}? This is now a finite integer-algebra problem on a fixed coefficient system.
2. Can a pooled coordinate be eliminated while all rows stay quadratic and the signed bootstrap remains valid? The two-row lemma is a construction, not a minimum theorem.
3. Can the local square-shell proof circuit be redesigned to retain more common quadratic directions through compilation, while preserving the constructor recoding and freshness guards?
4. Can low-degree consequences beyond constant row combinations and common Hessian kernels expose further integral projections? Rational eliminations need integral lifts or separate divisibility arguments.
5. A focused formalization could first establish Lemma 4.1 and Theorem 6.1, and then validate a compact certificate format. That isolates the new mathematics without requiring a new full logical build.

These questions are not assumptions of either delivered equation.

A Every residual of the principal equation

The following list fixes all coefficients of $P_{218} = \sum_{j=0}^{154} R_j^2$. The only independent unknowns are $x_0, \dots, x_{217} \in \mathbb{Z}$. There are no omitted constants, variable exponents, range predicates, or additional side conditions.

$$R_0 = x_0 - x_0\,x_2 - x_2 + x_7$$

$$R_1 = -x_2\,x_5 - x_3 + x_6\,x_7$$

$$R_2 = -1 + x_1 - x_6\,x_7$$

$$R_3 = 1 + x_3 - x_4\,x_{18} + 2x_{16}$$

$$R_4 = -1 + 2x_8 - x_8^2 - 2x_8\,x_9 + 2x_9 - x_9^2 + x_{19}$$

$$R_5 = -4 - 12x_8 - 16x_9 - 4x_{19} + x_{20} - x_{33}$$

$$R_6 = -2 + 2x_8 - x_8^2 - 4x_8\,x_{20} + 4x_{20} - 4x_{20}^2 + x_{21}$$

$$\begin{aligned} R_7 = 67 &- 60x_8 - 3x_8^2 - 60x_8\,x_{20} - 24x_8\,x_{21} - 116x_{20} - 124x_{20}^2 - 64x_{20}\,x_{21} \\ &- 31x_{21} - 4x_{21}^2 + x_{22} \end{aligned}$$

$$R_8 = -2 - x_9^2 - 2x_9\,x_{10} - x_{10}^2 + x_{23}$$

$$R_9 = -1 + 2x_8 - x_8^2 - 2x_8\,x_{10} + 2x_{10} - x_{10}^2 + x_{24}$$

$$\begin{aligned} R_{10} = 15 &- 38x_8 - 29x_8^2 - 34x_8\,x_9 - 36x_8\,x_{10} - 24x_8\,x_{19} - 24x_8\,x_{24} + 25x_9 + 38x_9^2 \\ &- 48x_9\,x_{10} + 16x_9\,x_{23} - 28x_9\,x_{24} + 36x_{10} + 70x_{10}^2 - 32x_{10}\,x_{19} + 32x_{10}\,x_{23} \\ &- 17x_{19} - 8x_{19}\,x_{24} + 40x_{23} - 14x_{24} + x_{25} - x_{29} - x_{33} \end{aligned}$$

$$\begin{aligned} R_{11} = -15 &+ 14x_8 - 2x_8^2 - 8x_8\,x_9 - 8x_8\,x_{23} + 4x_9^2 + 8x_9\,x_{23} + 4x_{23}^2 - 4x_{25}^2 \\ &+ x_{26} - 3x_{44} \end{aligned}$$

$$R_{12} = -6 + 16x_{10} - 16x_{10}^2 - 16x_{10}\,x_{23} - 16x_{10}\,x_{24} + 8x_{23} - 4x_{23}^2 - 8x_{23}\,x_{24} + 8x_{24} - 4x_{24}^2 + x_{27}$$

$$R_{13} = -6 - 48x_8\,x_{10} - 48x_8\,x_{24} - 24x_8\,x_{27} - 64x_9\,x_{10} - 64x_9\,x_{24} - 32x_9\,x_{27} + 7x_{10} + 9x_{10}^2 - 16x_{10}\,x_{19} + 8x_{10}\,x_{24} + 4x_{10}\,x_{27} - 16x_{19}\,x_{24} - 8x_{19}\,x_{27} + 2x_{20} + 8x_{24} + 4x_{27} + x_{28} - 3x_{33} - x_{43}$$

$$R_{14} = 6 - 19x_8 - 35x_8^2 - 80x_8\,x_{10} + 4x_8\,x_{23} - 24x_8\,x_{24} + 33x_9 + 35x_9^2 + 80x_9\,x_{10} + 24x_9\,x_{23} - 4x_9\,x_{24} + 16x_{10} + 32x_{10}\,x_{23} - 32x_{10}\,x_{24} + 12x_{23} + 4x_{23}^2 - 8x_{24} - 4x_{24}^2 + x_{29} + 4x_{29}^2 - x_{30}$$

$$R_{15} = 90 - 10x_8 - 2x_8^2 - 24x_8\,x_9 - 32x_8\,x_{10} - 8x_8\,x_{23} - 72x_9 - 18x_9^2 - 84x_9\,x_{10} - 48x_9\,x_{23} - 96x_{10} - 74x_{10}^2 - 64x_{10}\,x_{23} - 78x_{23} - 8x_{23}^2 - 4x_{29}^2 + x_{30}$$

$$R_{16} = -12 + 16x_8 - 4x_8^2 - 8x_8\,x_{10} - 8x_8\,x_{19} - 8x_8\,x_{23} + 16x_{10} - 4x_{10}^2 - 8x_{10}\,x_{19} - 8x_{10}\,x_{23} + 16x_{19} - 4x_{19}^2 - 8x_{19}\,x_{23} + 16x_{23} - 4x_{23}^2 + x_{31}$$

$$R_{17} = 25 - 56x_8 - 48x_8^2 - 128x_8\,x_{10} - 64x_8\,x_{23} - 32x_8\,x_{24} - 32x_8\,x_{31} - 90x_{10} - 52x_{10}^2 - 48x_{10}\,x_{23} - 8x_{10}\,x_{24} - 40x_{10}\,x_{31} - 24x_{23} - 8x_{23}^2 - 16x_{23}\,x_{31} - 38x_{24} + 4x_{24}^2 - 8x_{24}\,x_{31} - 2x_{27} - 4x_{31} - 4x_{31}^2 + x_{32}$$

$$R_{18} = 8 + 6x_8 - 8x_8^2 - 64x_8\,x_9 + 16x_8\,x_{19} - 64x_9^2 + 6x_{19} + 8x_{19}^2 - x_{20} + 3x_{33} - x_{37}$$

$$R_{19} = -2 + 2x_8 - x_8^2 - 4x_8\,x_{33} + 4x_{33} - 4x_{33}^2 + x_{34}$$

$$R_{20} = 67 - 60x_8 - 3x_8^2 - 60x_8\,x_{33} - 24x_8\,x_{34} - 116x_{33} - 124x_{33}^2 - 64x_{33}\,x_{34} - 31x_{34} - 4x_{34}^2 + x_{35}$$

$$R_{21} = -x_9^2 - 4x_9\,x_{35} - 4x_{35}^2 + x_{36}$$

$$R_{22} = -1 - 28x_8^2 - 32x_8\,x_9 - 40x_8\,x_{19} + 8x_9 - 32x_9\,x_{19} - 4x_{19} - 12x_{19}^2 - x_{20} + x_{37}$$

$$R_{23} = 49 - 34x_9^2 - 44x_9\,x_{10} - 24x_9\,x_{23} - 32x_9\,x_{37} + 6x_{10} - 10x_{10}^2 - 24x_{10}\,x_{23} - 24x_{10}\,x_{37} - 32x_{23} - 8x_{23}\,x_{37} + 4x_{37} - 4x_{37}^2 + x_{38} - x_{44}$$

$$R_{24} = 8x_8 - 64x_8^2 - 96x_8\,x_{10} - 32x_8\,x_{24} - 32x_8\,x_{38} + 6x_{10} - 36x_{10}^2 - 24x_{10}\,x_{24} - 24x_{10}\,x_{38} + 2x_{24} - 4x_{24}^2 - 8x_{24}\,x_{38} + 4x_{38} - 4x_{38}^2 + x_{39}$$

$$R_{25} = -x_{10}^2 - 2x_{10}\,x_{11} - x_{11}^2 + x_{40}$$

$$R_{26} = -11 + 3x_8^2 + 8x_8\,x_{10} - 8x_8\,x_{24} - 4x_8\,x_{40} + 4x_9^2 + 8x_9\,x_{23} - 8x_9\,x_{24} - 8x_{10} + 2x_{10}^2 - 8x_{10}\,x_{40} + 4x_{23}^2 - 8x_{23}\,x_{24} + 18x_{24} - 4x_{24}^2 + 4x_{25}^2 - x_{26} + 4x_{40} - 4x_{40}^2 + x_{41}$$

$$R_{27} = 4x_8 - 18x_8^2 - 40x_8\,x_{10} - 32x_8\,x_{24} - 40x_8\,x_{40} - 16x_8\,x_{41} + 4x_{10} - 24x_{10}^2 - 32x_{10}\,x_{24} - 48x_{10}\,x_{40} - 16x_{10}\,x_{41} - 16x_{24}^2 - 32x_{24}\,x_{40} - 16x_{24}\,x_{41} + 4x_{40} - 24x_{40}^2 - 16x_{40}\,x_{41} - 4x_{41}^2 + x_{42}$$

$$R_{28} = x_{10} - 25x_{10}^2 - 40x_{10}\,x_{24} - 20x_{10}\,x_{27} - 16x_{24}^2 - 16x_{24}\,x_{27} - 4x_{27}^2 + x_{43}$$

$$R_{29} = -1 + 8x_9 - 4x_9^2 - 8x_9\,x_{23} + 8x_{23} - 4x_{23}^2 + x_{44}$$

$$\begin{aligned} R_{30} = {} & -1 - 16x_9^2 - 8x_9\,x_{10} - 32x_9\,x_{23} - 16x_9\,x_{44} - x_{10}^2 - 8x_{10}\,x_{23} - 4x_{10}\,x_{44} \\ & - 16x_{23}^2 - 16x_{23}\,x_{44} - 4x_{44}^2 + x_{45} \end{aligned}$$

$$R_{31} = 3 + x_9 - 2x_9\,x_{10} - 4x_9\,x_{45} - 3x_{10}^2 - 8x_{10}\,x_{45} - x_{23} - 4x_{45}^2 + x_{46}$$

$$R_{32} = -1 - 4x_{46} - 4x_{46}^2 + x_{47}$$

$$R_{33} = -1 - x_{10}^2 - 8x_{10}\,x_{46} - 4x_{10}\,x_{47} - 16x_{46}^2 - 16x_{46}\,x_{47} - 4x_{47}^2 + x_{48}$$

$$R_{34} = 4 + x_9 - 2x_9\,x_{10} - 4x_9\,x_{48} - 3x_{10}^2 - 8x_{10}\,x_{48} - x_{23} - 4x_{48}^2 + x_{49}$$

$$R_{35} = -4x_{43}^2 - 8x_{43}\,x_{49} - 4x_{49}^2 + x_{50}$$

$$\begin{aligned} R_{36} = {} & -9 + 7x_8 - x_8^2 - 8x_8\,x_{43} - 4x_8\,x_{50} + 24x_{43} - 16x_{43}^2 - 16x_{43}\,x_{50} + 12x_{50} \\ & - 4x_{50}^2 + x_{51} \end{aligned}$$

$$R_{37} = -1 - 4x_{51} - 4x_{51}^2 + x_{52}$$

$$R_{38} = -x_{10}^2 - 8x_{10}\,x_{51} - 4x_{10}\,x_{52} - 16x_{51}^2 - 16x_{51}\,x_{52} - 4x_{52}^2 + x_{53}$$

$$R_{39} = -x_8 - x_8^2 - 4x_8\,x_{10} - 4x_8\,x_{53} - 4x_{10} - 4x_{10}^2 - 8x_{10}\,x_{53} - 4x_{53} - 4x_{53}^2 + x_{54}$$

$$R_{40} = -10 - 12x_{54} - 4x_{54}^2 + x_{55}$$

$$\begin{aligned} R_{41} = {} & 2 - 4x_9 - 16x_9\,x_{54} - 8x_9\,x_{55} - 4x_{23} - 16x_{23}\,x_{54} - 8x_{23}\,x_{55} - x_{44} + 8x_{54} \\ & - 16x_{54}^2 - 16x_{54}\,x_{55} + 4x_{55} - 4x_{55}^2 + x_{56} \end{aligned}$$

$$\begin{aligned} R_{42} = {} & -10 + x_9 - 25x_9^2 - 40x_9\,x_{23} - 20x_9\,x_{56} - 16x_{23}^2 - 16x_{23}\,x_{56} - 12x_{49} - 4x_{49}^2 \\ & - 4x_{56}^2 + x_{57} \end{aligned}$$

$$R_{43} = 21 + 24x_{49} - 8x_{49}^2 - 16x_{49}\,x_{57} - 2x_{57} - 4x_{57}^2 + x_{58}$$

$$\begin{aligned} R_{44} = {} & 1 + 4x_8 - 4x_8^2 - 8x_8\,x_{19} - 8x_8\,x_{60} + x_{10} - x_{10}^2 - 4x_{10}\,x_{58} + 4x_{19} - 4x_{19}^2 \\ & - 8x_{19}\,x_{60} - 4x_{58}^2 + x_{59} + 4x_{60} - 4x_{60}^2 \end{aligned}$$

$$R_{45} = -4x_8 - 4x_8^2 - 4x_8\,x_{10} - 8x_8\,x_{19} - x_{10} - x_{10}^2 - 4x_{10}\,x_{19} - 4x_{19} - 4x_{19}^2 + x_{60}$$

$$\begin{aligned} R_{46} = {} & 4x_8 - 24x_8^2 - 48x_8\,x_{19} - 16x_8\,x_{59} - 16x_8\,x_{60} + 4x_{19} - 24x_{19}^2 - 16x_{19}\,x_{59} \\ & - 16x_{19}\,x_{60} - 4x_{59}^2 + 8x_{60} - 8x_{60}^2 + x_{61} \end{aligned}$$

$$R_{47} = -2 - 4x_{61} - 4x_{61}^2 + x_{62}$$

$$R_{48} = -x_3\,x_{13} + x_{63}$$

$$R_{49} = x_{12} - x_{13} - 2x_{22}\,x_{63} - x_{64}$$

$$R_{50} = x_{12} - x_{13} - 2x_{26}\,x_{63} - x_{65}$$

$$R_{51} = -2x_{28}\,x_{63} + 2x_{42}\,x_{63} - 2x_{63} - x_{66} + x_{71}$$

$$R_{52} = x_{12} - x_{13} - 2x_{30}\,x_{63} - x_{67}$$

$$R_{53} = x_{12} - x_{13} - 2x_{32}\,x_{63} + 6x_{63} - x_{68}$$

$$R_{54} = x_{12} - x_{13} - 4x_{35}\,x_{63} - 2x_{36}\,x_{63} - x_{69}$$

$$R_{55} = x_{12} - x_{13} - 2x_{39}\,x_{63} - x_{70}$$

$$R_{56} = x_{12} - x_{13} - 2x_{42}\,x_{63} - 4x_{63} - x_{71}$$

$$R_{57} = -x_8\,x_9 + x_8\,x_{10} + x_9 + x_9\,x_{10} - x_{10} - x_{10}^2 + x_{72}$$

$$R_{58} = 1 + 2x_{11} + x_{11}^2 - x_{72}^2 + x_{73}$$

$$R_{59} = -x_{12}^2 + 2x_{12}\,x_{13} + 2x_{12}\,x_{75} - x_{13}^2 - 2x_{13}\,x_{75} - x_{73}^2 + x_{74} - x_{75}^2$$

$$R_{60} = -4x_{61}\,x_{63} - 2x_{62}\,x_{63} + 2x_{63} + x_{75}$$

$$R_{61} = -x_9\,x_{63} - x_{13} + x_{76}\,x_{77} + x_{78}$$

$$R_{62} = -1 + 2x_3 - 2x_3\,x_8 - 2x_3\,x_{19} + x_{76}$$

$$R_{63} = -x_8\,x_{213} - x_{12} + x_{77} + x_{213}$$

$$R_{64} = -6x_8\,x_{63} - 8x_9\,x_{63} + x_9\,x_{213} + x_{12} - x_{13} - 2x_{19}\,x_{63} + x_{79}$$

$$R_{65} = -x_{12}\,x_{64} + x_{13}\,x_{64} + x_{80}$$

$$R_{66} = -x_{65}\,x_{80} + x_{81}$$

$$R_{67} = -x_{66}\,x_{81} + x_{82}$$

$$R_{68} = -x_{67}\,x_{82} + x_{83}$$

$$R_{69} = -x_{68}\,x_{83} + x_{84}$$

$$R_{70} = -x_{69}\,x_{84} + x_{85}$$

$$R_{71} = -x_{70}\,x_{85} + x_{86}$$

$$R_{72} = -x_{71}\,x_{86} + x_{87}$$

$$R_{73} = -x_{78}\,x_{87} + x_{88}$$

$$R_{74} = -x_{79}\,x_{88} + x_{89}$$

$$R_{75} = -x_{89}^2 + x_{90}$$

$$R_{76} = -x_2 - 2x_2\,x_{16} + 2x_{16} + x_{91}$$

$$R_{77} = -x_{14}\,x_{91} + x_{92}$$

$$R_{78} = -2x_1^2 + 2x_1\,x_{12} + 2x_1\,x_{13} + 2x_1\,x_{92} + 2x_1\,x_{93} - x_{12}^2 - 2x_{12}\,x_{92} - x_{13}^2 - 2x_{13}\,x_{93} + x_{17}\,x_{95} - x_{74}\,x_{90} - x_{92}^2 - x_{93}^2$$

$$R_{79} = -x_2\,x_{15} + x_{15} - x_{15}\,x_{91} + x_{93}$$

$$R_{80} = -9x_8 - 9x_9 - 4x_{19} - 2x_{22} - 2x_{26} - 2x_{28} - 2x_{30} - 2x_{32} - 4x_{35} - 2x_{36} - 2x_{39} - 2x_{42} - 4x_{61} - 2x_{62} + x_{94}$$

$$R_{81} = -x_{96}\,x_{99} + x_{99} + x_{101}$$

$$R_{82} = -x_4\,x_{122} + x_{97}\,x_{204} - x_{100} - x_{101}\,x_{122}$$

$$R_{83} = x_{102}\,x_{204} - x_{103}\,x_{125} - x_{104}$$

$$R_{84} = x_{105}\,x_{204} - x_{106}\,x_{125} - x_{107}$$

$$R_{85} = x_{108}\,x_{204} - x_{109}\,x_{125} - x_{110}$$

$$R_{86} = x_{111}\,x_{204} - x_{112}\,x_{125} - x_{113}$$

$$R_{87} = x_{114}\,x_{205} - x_{116}\,x_{125} - x_{117}$$

$$R_{88} = x_{118}\,x_{205} - x_{119}\,x_{125} - x_{120}$$

$$R_{89} = -x_{17}\,x_{122} + x_{125}$$

$$R_{90} = x_{121} - x_{123}\,x_{125} - x_{124}$$

$$R_{91} = -x_{125}\,x_{127} + x_{126} - x_{128}$$

$$R_{92} = -x_{125}\,x_{130} + x_{129}\,x_{204} - x_{131}$$

$$R_{93} = -x_{125}\,x_{133} + x_{132}\,x_{204} - x_{134}$$

$$R_{94} = x_{135} - 2x_{135}\,x_{137} + x_{138}$$

$$R_{95} = -x_{135}\,x_{138} - x_{138} + x_{139}$$

$$R_{96} = 2x_{136} + x_{136}^2 + x_{138}^2 - x_{139}^2$$

$$R_{97} = -x_{136}\,x_{141} - x_{141} + x_{142}$$

$$R_{98} = -1 + x_{140}^2 + 2x_{140}\,x_{142} + x_{141}^2$$

$$R_{99} = 1 - 2x_0 + x_0^2 - 2x_0\,x_{136} + x_{143}$$

$$R_{100} = -x_0\,x_{141} + x_3 - x_{140} + x_{143}\,x_{144}$$

$$R_{101} = 1 - 2x_{4} + x_{4}^{2} - 2x_{4}\,x_{136} + x_{145}$$

$$R_{102} = -x_{4}\,x_{177} + x_{17} + x_{145}\,x_{146} - x_{176}$$

$$R_{103} = -1 + 2x_{136}\,x_{188} - x_{147} + 2x_{188} - x_{188}^{2}$$

$$R_{104} = -1 + x_{96} + x_{147}\,x_{148} - x_{176} - x_{177}\,x_{188}$$

$$R_{105} = -x_{136}\,x_{150} - x_{150} + x_{151}$$

$$R_{106} = -1 + x_{149}^{2} + 2x_{149}\,x_{151} + x_{150}^{2}$$

$$R_{107} = 1 + x_{136} - x_{136}\,x_{153} - x_{153} + x_{154}$$

$$R_{108} = x_{152}^{2} + 2x_{152}\,x_{154} - 2x_{153} + x_{153}^{2}$$

$$R_{109} = -x_{136}\,x_{156} - x_{156} + x_{157}$$

$$R_{110} = -1 + x_{155}^{2} + 2x_{155}\,x_{157} + x_{156}^{2}$$

$$R_{111} = -x_{136}\,x_{159} - x_{159} + x_{160}$$

$$R_{112} = -1 + x_{158}^{2} + 2x_{158}\,x_{160} + x_{159}^{2}$$

$$R_{113} = -x_{136}\,x_{162} - x_{162} + x_{163}$$

$$R_{114} = -1 + x_{161}^{2} + 2x_{161}\,x_{163} + x_{162}^{2}$$

$$R_{115} = -x_{136}\,x_{165} - x_{165} + x_{166}$$

$$R_{116} = -1 + x_{164}^{2} + 2x_{164}\,x_{166} + x_{165}^{2}$$

$$R_{117} = -x_{136}\,x_{168} - x_{168} + x_{169}$$

$$R_{118} = -1 + x_{167}^{2} + 2x_{167}\,x_{169} + x_{168}^{2}$$

$$R_{119} = -x_{136}\,x_{171} - x_{171} + x_{172}$$

$$R_{120} = -1 + x_{170}^{2} + 2x_{170}\,x_{172} + x_{171}^{2}$$

$$R_{121} = -x_{136}\,x_{174} - x_{174} + x_{175}$$

$$R_{122} = -1 + x_{173}^{2} + 2x_{173}\,x_{175} + x_{174}^{2}$$

$$R_{123} = 1 - x_{136}\,x_{177} - x_{177} + x_{178}$$

$$R_{124} = -1 + 2x_{176} + x_{176}^{2} + 2x_{176}\,x_{178} + x_{177}^{2}$$

$$R_{125} = -x_{136}\,x_{180} - x_{180} + x_{181}$$

$$R_{126} = -1 + x_{179}^2 + 2x_{179}\,x_{181} + x_{180}^2$$

$$R_{127} = -x_{136}\,x_{183} - x_{183} + x_{184}$$

$$R_{128} = -1 + x_{182}^2 + 2x_{182}\,x_{184} + x_{183}^2$$

$$R_{129} = -x_{136}\,x_{186} - x_{186} + x_{187}$$

$$R_{130} = -1 + x_{185}^2 + 2x_{185}\,x_{187} + x_{186}^2$$

$$\begin{aligned} R_{131} = {} & -3 - 2x_0 - 4x_0^2 - 2x_0\,x_3 - x_1^2 - x_2^2 - 2x_3^2 - x_4^2 + 2x_8 - x_8^2 + 2x_9\,x_{10} \\ & - x_{11}^2 - x_{12}^2 - x_{13}^2 - x_{14}^2 - x_{15}^2 - 4x_{16}^2 - x_{17}^2 - x_{23} - x_{94}^2 + 2x_{96} \\ & - 2x_{96}^2 - x_{97}^2 - x_{98}^2 - x_{102}^2 - x_{105}^2 - x_{108}^2 - x_{111}^2 - x_{114}^2 - x_{115}^2 \\ & - x_{118}^2 - x_{121}^2 - x_{122}^2 - x_{126}^2 - x_{129}^2 - x_{132}^2 + x_{135} - x_{188}^2 \end{aligned}$$

$$\begin{aligned} R_{132} = {} & 2x_4 - 4x_4\,x_8 - 4x_4\,x_9 - 4x_4\,x_{10} - 4x_4\,x_{11} - 4x_4\,x_{12} - 4x_4\,x_{13} - 4x_4\,x_{14} \\ & - 4x_4\,x_{15} - 8x_4\,x_{16} + x_{188} \end{aligned}$$

$$R_{133} = -16 + x_0 - 4x_{189}^2$$

$$R_{134} = -x_{136}\,x_{191} - x_{191} + x_{200}$$

$$R_{135} = -1 + x_{190}^2 + x_{191}^2 - x_{200}^2$$

$$R_{136} = -x_{193} - x_{193}\,x_{194} + x_{201}$$

$$R_{137} = -1 + x_{192}^2 + x_{193}^2 - x_{201}^2$$

$$R_{138} = -x_{186}^2 + x_{202}$$

$$R_{139} = x_{191} - 2x_{196}\,x_{202} - x_{202}$$

$$R_{140} = -4x_{191} - 8x_{191}\,x_{195} + x_{194}$$

$$R_{141} = -x_{136} - x_{190}\,x_{197} + x_{194}$$

$$R_{142} = -x_{185} - x_{187} - x_{190}\,x_{198} + x_{192}$$

$$R_{143} = -x_{94} - 4x_{186}\,x_{199} + x_{193}$$

$$\begin{aligned} R_{144} = {} & 26 + 2x_0^2 + 2x_0\,x_3 - 2x_0\,x_{141} - 2x_0\,x_{150} + x_1^2 - 2x_1\,x_{177} + x_2^2 - 2x_2\,x_{168} \\ & + x_3^2 - 2x_3\,x_{150} + x_8^2 - 2x_8\,x_{153} + x_9^2 - 2x_9\,x_{156} + x_{10}^2 - 2x_{10}\,x_{159} + x_{11}^2 \\ & - 2x_{11}\,x_{162} + x_{12}^2 - 2x_{12}\,x_{165} + x_{13}^2 - 2x_{13}\,x_{171} + x_{14}^2 - 2x_{14}\,x_{174} + x_{15}^2 \\ & - 2x_{15}\,x_{180} + 4x_{16}^2 - 4x_{16}\,x_{183} + x_{94}^2 - 2x_{94}\,x_{186} - x_{136}\,x_{203} - 2x_{140} - 2x_{142} \\ & + x_{142}^2 - 2x_{149} - 2x_{151} + x_{151}^2 - 2x_{152} + 2x_{153} - 2x_{154} + x_{154}^2 - 2x_{155} \\ & - 2x_{157} + x_{157}^2 - 2x_{158} - 2x_{160} + x_{160}^2 - 2x_{161} - 2x_{163} + x_{163}^2 - 2x_{164} \\ & - 2x_{166} + x_{166}^2 - 2x_{167} - 2x_{169} + x_{169}^2 - 2x_{170} - 2x_{172} + x_{172}^2 - 2x_{173} \\ & - 2x_{175} + x_{175}^2 - 2x_{176} + x_{178}^2 - 2x_{179} - 2x_{181} + x_{181}^2 - 2x_{182} - 2x_{184} \\ & + x_{184}^2 - 2x_{185} - 2x_{187} + x_{187}^2 \end{aligned}$$

$$R_{145} = -x_{98}\,x_{204} + x_{122}$$

$$R_{146} = -x_{115}\,x_{205} + x_{122}$$

$$R_{147} = -1 + x_2 - x_3 - 2x_3\,x_{212}$$

$$\begin{aligned} R_{148} = {} & -2 - x_{94}^2 - x_{141}^2 - x_{150}^2 + 2x_{153} - x_{153}^2 - x_{156}^2 - x_{159}^2 - x_{162}^2 - x_{165}^2 \\ & - x_{168}^2 - x_{171}^2 - x_{174}^2 - x_{177}^2 - x_{180}^2 - x_{183}^2 + x_{186}^2 - x_{206}^2 - x_{207}^2 - x_{208} \\ & - x_{208}^2 \end{aligned}$$

$$\begin{aligned} R_{149} = {} & -1 - 4x_{100}^2 - 4x_{104}^2 - 4x_{107}^2 - 4x_{110}^2 - 4x_{113}^2 - 4x_{117}^2 - 4x_{120}^2 \\ & + x_{122}^2 - 4x_{124}^2 - 4x_{128}^2 - 4x_{131}^2 - 4x_{134}^2 - x_{209}^2 - x_{210}^2 - x_{211} - x_{211}^2 \end{aligned}$$

$$R_{150} = -x_3\,x_{12} + x_{213}$$

$$\begin{aligned} R_{151} = {} & x_{96}\,x_{215} - 2x_{97}\,x_{150} + 2x_{102} - 2x_{102}\,x_{153} - 2x_{105}\,x_{156} - 2x_{108}\,x_{159} - 2x_{111}\,x_{162} \\ & - 2x_{114}\,x_{165} - 2x_{118}\,x_{171} - 2x_{121}\,x_{174} - 2x_{126}\,x_{180} - 2x_{129}\,x_{183} - 2x_{132}\,x_{186} \\ & - 2x_{136}\,x_{215} + 2x_{149}\,x_{150} + 2x_{150}\,x_{151} - 2x_{152} + 2x_{152}\,x_{153} + 2x_{153}\,x_{154} - 2x_{154} \\ & + 2x_{155}\,x_{156} + 2x_{156}\,x_{157} + 2x_{158}\,x_{159} + 2x_{159}\,x_{160} + 2x_{161}\,x_{162} + 2x_{162}\,x_{163} \\ & + 2x_{164}\,x_{165} + 2x_{165}\,x_{166} + 2x_{170}\,x_{171} + 2x_{171}\,x_{172} + 2x_{173}\,x_{174} + 2x_{174}\,x_{175} \\ & + 2x_{179}\,x_{180} + 2x_{180}\,x_{181} + 2x_{182}\,x_{183} + 2x_{183}\,x_{184} + 2x_{185}\,x_{186} + 2x_{186}\,x_{187} \\ & + x_{214} - 2x_{215} \end{aligned}$$

$$\begin{aligned} R_{152} = {} & 1 + x_{94}^2 - x_{96}\,x_{214} + x_{97}^2 - 2x_{97}\,x_{149} + x_{102}^2 - 2x_{102}\,x_{152} + x_{105}^2 \\ & - 2x_{105}\,x_{155} + x_{108}^2 - 2x_{108}\,x_{158} + x_{111}^2 - 2x_{111}\,x_{161} + x_{114}^2 - 2x_{114}\,x_{164} \\ & + x_{118}^2 - 2x_{118}\,x_{170} + x_{121}^2 - 2x_{121}\,x_{173} + x_{126}^2 - 2x_{126}\,x_{179} + x_{129}^2 \\ & - 2x_{129}\,x_{182} + x_{132}^2 - 2x_{132}\,x_{185} + x_{141}^2 + x_{149}^2 + x_{152}^2 + x_{155}^2 + x_{158}^2 \\ & + x_{161}^2 + x_{164}^2 + x_{168}^2 + x_{170}^2 + x_{173}^2 + x_{177}^2 + x_{179}^2 + x_{182}^2 + x_{185}^2 \\ & - 2x_{186}^2 + x_{206}^2 + x_{207}^2 + x_{208} + x_{208}^2 + x_{215} \end{aligned}$$

$$\begin{aligned} R_{153} = {} & x_{96}\,x_{217} - 2x_{98}\,x_{141} - 2x_{115}\,x_{168} - 2x_{122}\,x_{177} - 2x_{136}\,x_{217} + 2x_{140}\,x_{141} \\ & + 2x_{141}\,x_{142} + 2x_{167}\,x_{168} + 2x_{168}\,x_{169} + 2x_{176}\,x_{177} + 2x_{177} + 2x_{177}\,x_{178} + x_{216} \\ & - 3x_{217} \end{aligned}$$

$$\begin{aligned} R_{154} = {} & -x_{96}\,x_{216} + x_{98}^2 - 2x_{98}\,x_{140} + x_{115}^2 - 2x_{115}\,x_{167} + x_{122}^2 - 2x_{122}\,x_{176} + x_{140}^2 \\ & - x_{141}^2 + x_{167}^2 - x_{168}^2 + x_{176}^2 - x_{177}^2 + x_{216} + x_{217} \end{aligned}$$

B Artifact identities

Pinned 242-coordinate source. SHA-256:

`50f88c245d3a32b5c041924ea4ac07b9149e5da24c859f3a167ea984d93e18d2`

Composed 223-coordinate parent. SHA-256:

`ebf82c1646ff38125aaed2bdef0e7fc5c07b67853dda0e781fc459ead0d1e7c4`

Principal residual system. SHA-256:

`95c0db6befd4ad98e6749ba18580f3dd775b1faaec36d270f8b839911754d6b1`

Companion residual system. SHA-256:

`a91308bd0c82182ea15080763cb23cb4fa0aa45a812eabb7a6cbcd21c0b95a06`

Principal collected quartic. SHA-256:

`a3ee776477f2b6ac5eca2e5e55726e916f596317170d4d92452a2363504c400a`

Companion collected quartic. SHA-256:

`e5f7a57354358df48a06d9b82ff1bafc3baecde8ce927b1a504c7c015b79ccf9`

The algebra certificates contain the five radical directions, both affine combinations, the centered chart and its inverse, and every row operation. The source-composition certificates contain the cofinal-range edits, private-wire maps, pooled blocks, and complete frame-atom ledger.

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[2] ProveIt contributors. *Pooling exact power certificates for Diophantine consistency; source 158*. Same pinned snapshot. https://github.com/VladimirReshetnikov/ProveIt/tree/953fb4d82d4229f4efc5f9db4cc2bf38ae329bb4/Computability/DiophantineConsistency/con-zfc-diophantine-forms/158-pooled-powers.

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[6] ProveIt contributors. *A complete fifteen-variable consistency equation*. Complete display at the same snapshot. https://github.com/VladimirReshetnikov/ProveIt/blob/953fb4d82d4229f4efc5f9db4cc2bf38ae329bb4/Computability/DiophantineConsistency/equation.tex.

[7] ProveIt contributors. *Sources 160–165: later direct quartic and compact tradeoffs*. README comparison at commit `fb1bed5eb7196cf35abfaa746c18c6ee7eae6cfb`. https://github.com/VladimirReshetnikov/ProveIt/tree/fb1bed5eb7196cf35abfaa746c18c6ee7eae6cfb/Computability/DiophantineConsistency/con-zfc-diophantine-forms.